Capítulo 3

Base inversa

Tabla 3-1

c_i	A_i	20 A_1	45 A_2	0 A_3	0 A_4	0 A_5	x_i	x_i/y_{ij} ($y_{ij}>0$)
0	A_3	1	2	1	0	0	40	20
0	A_4	3	3/2	0	1	0	75	50
0	A_5	0	1	0	0	1	15	15
z_j		0	0	0	0	0	$z = 0$	
$c_j - z_j$		20	45	0	0	0		

A3	A4	A2
1	0	2
0	1	1.5
0	0	1

Inversa		
1	0	-2
0	1	-1.5
0	0	1

Tabla 3-2

c_i	A_i	20 A_1	45 A_2	0 A_3	0 A_4	0 A_5	x_i	x_i/y_{ij} ($y_{ij}>0$)
0	A_3	1	0	1	0	–2	10	10
0	A_4	3	0	0	1	–3/2	105/2	35/2
45	A_2	0	1	0	0	1	15	
z_j		0	45	0	0	45	$z = 675$	
$c_j - z_j$		20	0	0	0	–45		

A1	A4	A2
1	0	2
3	1	1.5
0	0	1

Inversa		
1	0	-2
-3	1	4.5
0	0	1

Tabla 3-3

c_i	A_i	20 A_1	45 A_2	0 A_3	0 A_4	0 A_5	x_i	x_i/y_{ij} ($y_{ij}>0$)
20	A_1	1	0	1	0	–2	10	
0	A_4	0	0	–3	1	9/2	45/2	
45	A_2	0	1	0	0	1	15	
z_j		20	45	20	0	5	$z = 875$	
$c_j - z_j$		0	0	–20	0	–5		

Coeficientes de sustitución y costos reducidos

$Ax = b$

$Bx_B + Nx_N = b$ $\quad x_{B,N} \geq 0$

$Bx_B = b - Nx_N$

$x_B = B^{-1}(b - Nx_N)$

$x_B = B^{-1}b - B^{-1}Nx_N$

$B^{-1}N = Y$ (ver factibilidad)

$x_B = B^{-1}b - Yx_N$

$x_B = B^{-1}b - \sum_j Y_j x_j$

$\frac{\partial x_B}{\partial x_j} = 0 - \sum_j Y_j$

$\frac{\partial x_1}{\partial x_j} = -Y_{1j}$; $\frac{\partial x_2}{\partial x_j} = -Y_{2j}$ → coef de sustitucion

Tabla 3-1

c_i	A_i	20 A_1	45 A_2	0 A_3	0 A_4	0 A_5	x_i	x_i/y_{ij} ($y_{ij}>0$)
0	A_3	1	2	1	0	0	40	20
0	A_4	3	3/2	0	1	0	75	50
0	A_5	0	1	0	0	1	15	15
z_j		0	0	0	0	0	$z = 0$	
$c_j - z_j$		20	45	0	0	0		

Tabla 3-2

c_i	A_i	20 A_1	45 A_2	0 A_3	0 A_4	0 A_5	x_i	x_i/y_{ij} ($y_{ij}>0$)
0	A_3	1	0	1	0	–2	10	10
0	A_4	3	0	0	1	–3/2	105/2	35/2
45	A_2	0	1	0	0	1	15	
z_j		0	45	0	0	45	$z = 675$	
$c_j - z_j$		20	0	0	0	–45		

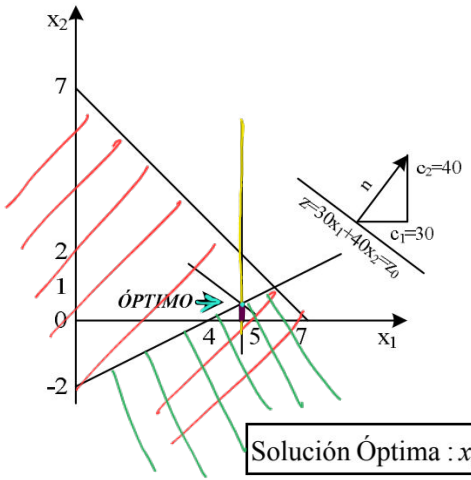
$z' = z + \theta_j (c_j - z_j)$ → ver factibilidad

$z' = z + x_j (c_j - z_j)$ → ver paso a paso del simplex

$\frac{\partial z}{\partial x_j} = 0 + c_j - z_j = c_j - z_j$

COSTO REDUCIDO

Variables ficticias y penalización



Solución Óptima : $x_1^* = 5,\ x_2^* = 1/2,\ z^* = 170$

$$\text{Max } z = 30x_1 + 40x_2$$

s. a

$$x_1 + x_2 \le 7$$
$$x_1 - 2x_2 \ge 4$$
$$x_1 = 5$$
$$x_1; x_2 \ge 0$$

$$\text{Max } z = 30x_1 + 40x_2 + 0x_3 + 0x_4$$

s. a

$$x_1 + x_2 + 1x_3 + 0x_4 = 7$$
$$x_1 - 2x_2 + 0x_3 - 1x_4 = 4$$
$$x_1 + 0x_2 + 0x_3 + 0x_4 = 5$$
$$x_j \ge 0\ j = 1, 2, \ldots, 4$$

ESTÁNDAR

MUY GRANDES

$$\text{Max } z = 30x_1 + 40x_2 + 0x_3 + 0x_4 - Mx_5 - Mx_6$$

MAX - MIN +

s. a

$$x_1 + x_2 + 1x_3 + 0x_4 + \mathbf{0x_5} + \mathbf{0x_6} = 7$$
$$x_1 - 2x_2 + 0x_3 - 1x_4 + \mathbf{1x_5} + \mathbf{0x_6} = 4$$
$$x_1 + 0x_2 + 0x_3 + 0x_4 + \mathbf{0x_5} + \mathbf{1x_6} = 5$$
$$x_j \ge 0\ j = 1, 2, \ldots, 6$$

VAR. FICTICIAS

Variables ficticias y penalización

$$\text{Max } z = 30x_1 + 40x_2 + 0x_3 + 0x_4 - Mx_5 - Mx_6$$

s. a

$$x_1 + x_2 + 1x_3 + 0x_4 + \mathbf{0x_5} + \mathbf{0x_6} = 7$$
$$x_1 - 2x_2 + 0x_3 - 1x_4 + \mathbf{1x_5} + \mathbf{0x_6} = 4$$
$$x_1 + 0x_2 + 0x_3 + 0x_4 + \mathbf{0x_5} + \mathbf{1x_6} = 5$$
$$x_j \geq 0 \; j = 1, 2, \ldots, 6$$

		30	40	0	0	–M	–M		x_i/y_{ij}
c_i	A_i	A_1	A_2	A_3	A_4	A_5	A_6	x_i	$(y_{ij}>0)$
0	A_3	1	1	1	0	0	0	7	7
–M	A_5	1	–2	0	–1	1	0	4	4
–M	A_6	1	0	0	0	0	1	5	5
z_j		–2M	2M	0	M	–M	–M	z = –9M	
$c_i - z_j$		30+2M	40–2M	0	–M	0	0		

		30	40	0	0	–M	–M		x_i/y_{ij}
c_i	A_i	A_1	A_2	A_3	A_4	A_5	A_6	x_i	$(y_{ij}>0)$
0	A_3	0	3	1	1	–1	0	3	1
30	A_1	1	–2	0	–1	1	0	4	
–M	A_6	0	2	0	1	–1	1	1	1/2
z_j		30	–60–2M	0	–30–M	30+M	–M	z = 120–M	
$c_i - z_j$		0	100+2M	0	30+M	–30–2M	0		

		30	40	0	0	–M	–M		x_i/y_{ij}
c_i	A_i	A_1	A_2	A_3	A_4	A_5	A_6	x_i	$(y_{ij}>0)$
0	A_3	0	0	1	–1/2	1/2	–3/2	3/2	
30	A_1	1	0	0	0	0	1	5	
40	A_2	0	1	0	1/2	–1/2	1/2	1/2	
z_j		30	40	0	20	–20	50	z = 170	
$c_i - z_j$		0	0	0	–20	20–M	–50–M		